Introduction to Topology

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Abstract

This course was taught by Dr. Yu-Shen Lin at Boston University during the Spring 2025 semester. We used the text Introduction to Topology, Pure and Applied by Colin Adams and Robert Franzosa.

Contents

Chapter 1

Topological Spaces

Lecture 1: Syllabus Day

Tue 21 Jan 2025 11:01

Textbook not required but I already bought it wooo also biweekly homework on blackboard always due at 11:59pm. We are encouraged to type homework (awewsomeeee) also why does this one guy in the class look like a literal reddit mod LOL also karim showed up 7m late. "I cannot give everyone an A" great its competitive now

"You've probably heard that a mug and a donut are the same thing" I love this class. Making this notion precise and rigorous is difficult, and by the end of the semester we will be able to prove this notion.

Definition 1.1: Topological Space

Let $X \neq \varnothing$ be a set. A topology on X is a collection of subsets $\mathcal{T} \subseteq \mathscr{P}(X)$ such that

1. $\varnothing, X \in \mathcal{T}$;
2. $\mathcal{T}$ is closed under *finite* intersection;
3. $\mathcal{T}$ is closed under union.

We call the pair $(X, \mathcal{T})$ a *topological space*.

Definition 1.2: Open Set

Let $(X, \mathcal{T})$ be a topological space. Any subset $A \subseteq X$ such that $A \in \mathcal{T}$ is called an *open set*.

Note that there is a finite restriction on (2), but not on (3). We will discuss this later. Definiton 1.1 is extremely abstract and seemingly arbitrary, so consider the following examples to build some intuition.

The most basic example is that of the *standard topology on* $\mathbb{R}$. Define $\mathcal{T}$ as the collection of all unions of open intervals of the form $(a, b) \subseteq \mathbb{R}$, together with the empty set. Trivially this definition satisfies property (1), since $(-\infty, \infty) = \mathbb{R}$ is open, and we define $\varnothing \in \mathcal{T}$. It also satisfies (3) by our definition. Showing (2) is slightly more difficult and requires ε-δ to make precise, and we will cover it later when we have this machinery.

Consider the open sets of the form

$$\left(\frac{-1}{2^n}, \frac{1}{2^n}\right), \quad n \in \mathbb{N}$$

in the standard topology on $\mathbb{R}$. However, if we take the intersection

$$\bigcap_{n \in \mathbb{N}} \left(-\frac{1}{2^n}, \frac{1}{2^n}\right) = \{0\}.$$

This is *not an open set* in the standard topology on $\mathbb{R}$, as it is not an open interval and not empty. This is why the finite criterion on (2) is important.

Consider a more simple example. On any set X, the trivial topology $\mathcal{T} = \{\varnothing, X\}$ is a topology. We trivially have criteria (1). Consider a finite collection $\{A_n\}_{n=0}^k$. There are two cases. If $A_n = \varnothing$ for some n, then

$$\bigcap_{n=0}^{k} A_n = \varnothing.$$

Otherwise, we have

$$\bigcap_{n=0}^{k} A_n = X,$$

since we must then have $A_n = X \forall n$. Since $\varnothing, X \in \mathcal{T}$, we have (2). The same argument applies for (3); if $X \in \{A_n\}_{n=0}^k$, we have the union is X; otherwise we have the union is empty. The trivial topology is also called the minimal topology, as it is the smallest possible topology since it must contain these subsets.

The next example is the discrete topology. Let $\mathcal{T} = \mathscr{P}(X)$ for some set X. This very trivially satisfies definition 1.1, and we will explain later why we call it the discrete topology. This is also called the maximal topology, since it is the largest possible topology.

Let $X = \mathbb{R}$. We will now define what is called the Zariski topology, or sometimes the finite compliment topology. Let $\mathcal{T}$ be defined as all sets with either finite compliment in $\mathbb{R}$, or the empty set:

$$\mathcal{T} \coloneqq \{Y \subseteq \mathbb{R} \mid Y = \varnothing \vee |\mathbb{R} \setminus Y| < \infty\}.$$

We will show this is a topology. By definition, $\varnothing \in \mathcal{T}$, and we have $\mathbb{R} \setminus \mathbb{R} = \varnothing$ is finite, so $\mathbb{R} \in \mathcal{T}$. The second example is more interesting. Consider a finite collection $\{A_n\}_{n=1}^k \subseteq \mathcal{T}$. We want to show

$$\bigcap_{i=1}^{k} A_i$$

is open. By our definition, a set is open if and only if it's compliment in $\mathbb{R}$ is finite. DeMorgan's Law states that

$$\left(\bigcap_{i=1}^{k} A_i\right)^{\complement} = \bigcup_{i=1}^{k} A_i^{\complement}.$$

"I assume you learned this in high school". Note that the maximum cardinality of this union occurs when $A_i \cap A_j = \varnothing$ for all $i \neq j$, so we have

$$\left|\bigcup_{i=1}^{k} A_i^{\complement}\right| = \sum_{i=1}^{k} \left|A_i^{\complement}\right|.$$

A finite sum of finite numbers is finite. More generally, a finite union of finite sets is finite. Now we show the third criterion holds. Suppose $\{A_i\}_{i\in I} \subseteq \mathcal{T}$. By definition, $\left|A_i^{\complement}\right| < \infty$ for all $i \in I$. We wish to show

$$\left(\bigcup_{i\in I} A_i\right)^{\complement} < \infty.$$

DeMorgan's Law also states that

$$\left(\bigcup_{i\in I} A_i\right)^{c} = \bigcap_{i\in I} A_i^{c}.$$

Since the intersection of finite sets is finite, we have that $\cup\{A_i\}_{i\in I}$ is open. The Zariski topology is the topology used in algebraic geometry.

Definition 1.3: Neighborhood

Let X be a topological space and let $x \in X$. An open set U such that $x \in U$ is called a *neighborhood* of x.

The intuition behind a neighborhood is the collection of points near x. This may not be true in a general topology, but it is true in many topologies.

Theorem 1.1

Let X be a topological space. Then $A \subseteq X$ is open if and only if for all $x \in A$, there exists a neighborhood U such that $x \in U \subseteq A$.

Proof. Suppose A is open and let $x \in A$. Then $U = A$ is a neighborhood of x. Now suppose for all $x \in A$, there exists an open neighborhood $U_x \subseteq A$ of x. Note that

$$A \subseteq \bigcup_{x \in A} U_x$$

because

$$y \in A \implies \exists U_y \text{ s.t. } y \in U_y \implies y \in \bigcup_{x \in A} U_x.$$

However, since every $U_x \subseteq A$, we have also that

$$\bigcup_{x \in A} U_x \subseteq A$$

since

$$y \in \bigcup_{x \in A} U_x \implies \exists U_x \text{ s.t. } y \in U_x \subseteq A.$$

Therefore, A is open by criteria (3) in definition 1.1. ■

Lecture 2: hi

Thu 23 Jan 2025 10:58

I walked in and we're talking about some stuff with boats and physics? I don't really know what's happening, but we did some vector stuff to conclude the best way to sail against the wind is at a 30 degree angle.

It is usually too difficult to list all open sets in a given topology, so we use a basis instead.

Definition 1.4: Basis

Let X be a nonempty set and let $\mathcal{B}$ be a collection of subsets of X. Then $\mathcal{B}$ is a *basis* of some topology $\mathcal{T}_{\mathcal{B}}$ on X iff

1. For all $x \in X$, there exists $V \in \mathcal{B}$ such that $x \in V$;

2. For all $V_1, V_2 \in \mathcal{B}$ with $x \in V_1 \cap V_2$, there exists $V_3 \in \mathcal{B}$ such that $x \in V_3 \subseteq V_1 \cap V_2$.

Proposition 1.1

Let $\mathcal{B}$ be a basis for some set X. Then the set $\mathcal{T}_{\mathcal{B}}$ of the empty set, together with all unions of elements of $\mathcal{B}$, forms a topology on X.

Proof. Recall the three axioms from definition 1.1. By definition, $\varnothing \in \mathcal{T}_{\mathcal{B}}$. By property (1) of definition 1.4, for all $x \in X$, there exists $V_x \in \mathcal{B}$ such that $x \in V_x$. By definition of $\mathcal{T}_{\mathcal{B}}$,

$$X = \bigcup_{x \in X} V_x \in \mathcal{T}_{\mathcal{B}}.$$

Therefore, property (1) is satisfied.

Now, we show property (2); closure under finite intersection. Let $\{A_i\}_{i=1}^k \subseteq \mathcal{T}_{\mathcal{B}}$. By definition, we have that A_i is either empty, or can be represented as a union

$$A_i = \bigcup_{j \in J_i} V_{ij}$$

for $\{V_{ij}\}_{j \in J_i} \subseteq \mathcal{B}$. If any $A_i = \varnothing$, we have that the intersection

$$\bigcap_{i=1}^k A_i = \varnothing \in \mathcal{T}_{\mathcal{B}}.$$

Thus, we can assume WLOG that $A_i \neq \varnothing$ for all i. We have

$$\bigcap_{i=1}^k A_i = \bigcap_{i=1}^k \bigcup_{j \in J_i} V_{ij}.$$

Recall that the distributive property says $a \cup (b \cap c) = (a \cap b) \cup (a \cap c)$. We can thus rewrite something along the lines of

$$\bigcap_{i=1}^k \bigcup_{j \in J_i} V_{ij} = \bigcup_{j_i \in J_i} \bigcap_{i \in I} V_{ij_i}.$$

We thus need only prove this intersection is an element of $\mathcal{T}_{\mathcal{B}}$. We prove this by strong induction. For $k = 1$, we have $V_1 = \bigcap_{i=1}^k B_i \in \mathcal{B} \subseteq \mathcal{T}_{\mathcal{B}}$. Now assuming the inductive hypothesis for $n \leq k$, we have for any $\{V_i\}_{i=1}^{k+1} \subseteq \mathcal{B}$, the intersection

$$\bigcap_{i=1}^k V_i \in \mathcal{B}.$$

We can thus represent it as a union of elements $\{V_i'\}_{i \in I} \in \mathcal{T}_{\mathcal{B}}$. By distributivity, it follows that

$$\bigcap_{i=1}^{k+1} V_i = \left(\bigcup_{i \in I} V_i' \right) \cap V_{k+1} = \bigcup_{i \in I} \left(V_i' \cap V_{k+1} \right).$$

By definition 1.4, for all i and for all $x \in V_i' \cap V_{k+1}$, there exists $V_x \subseteq V_i' \cap V_{k+1}$ such that $x \in V_x$. We then have

$$\bigcup_{i \in I} \left(V_i' \cap V_{k+1} \right) = \bigcup_{i \in I} \bigcup_{x \in V_i' \cap V_{k+1}} V_x.$$

Since $V_x \in \mathcal{B}$ for all x, by definition of $\mathcal{T}_{\mathcal{B}}$, we have

$$\bigcup_{i\in I} \bigcup_{x\in V_i' \cap V_{k+1}} V_x \in \mathcal{T}_{\mathcal{B}}.$$

By induction, we are done.

Finally, we show property (3). Let $\{A_i\}_{i\in I} \subseteq \mathcal{T}_{\mathcal{B}}$. By definition of $\mathcal{T}_{\mathcal{B}}$, for all i there exists $\{B_{ij}\}_{j\in J_i}$ such that

$$A_i = \bigcup_{j\in J_i} B_{ij}.$$

We then have

$$\bigcup_{i\in I} A_i = \bigcup_{i\in I} \bigcup_{j\in J_i} B_{ij} = \bigcup_{i\in I,\, j\in J_i} B_{ij}.$$

Since this is a union of elements in $\mathcal{B}$, by definition of $\mathcal{T}_{\mathcal{B}}$, we have that

$$\bigcup_{i\in I,\, j\in J_i} B_{ij} \in \mathcal{T}_{\mathcal{B}}.$$

■

Definition 1.5: Topology Generated by a Basis

Let $\mathcal{B}$ be a basis for a set X. Then the set $\mathcal{T}_{\mathcal{B}}$ consisting of the empty set together with all unions of elements in $\mathcal{B}$ is a topology on X, called the *topology generated by* $\mathcal{B}$.

For example, define

$$B_r(x) \coloneqq \{a \in \mathbb{R} \mid |x-a| < r\}.$$

Then the collection

$$\mathcal{B} = \left\{B_r(x) \mid x \in \mathbb{R}, r \in \mathbb{R}^+\right\}$$

is a basis for the standard topology on $\mathbb{R}$. This generalizes; let

$$B_r(x) \coloneqq \{a \in \mathbb{R}^n \mid \|x-a\| < r\}.$$

Then

$$\mathcal{B} = \left\{B_r(x) \mid r \in \mathbb{R}^+, x \in \mathbb{R}^n\right\}$$

is a basis for $\mathbb{R}^n$. We will prove this. Let $x \in \mathbb{R}^n$. Note that for any choice $r \in \mathbb{R}^+$, we have $x \in B_r(x)$. This shows property (1) of definition 1.4. As for property (2), let $x \in B_{r_2}(x_1) \cap B_{r_2}(x_2)$. It follows that this intersection is nonempty, which implies that $\|x_1 - x\|, \|x_2 - x\| < r_1, r_2$. We wish to find some r such that $B_r(x) \subseteq B_{r_1}(x_1) \cap B_{r_2}(x_2)$. I propose we choose $|r_1 - r_2|$. Let $y \in B_{|r_1-r_2|}(x)$. We wish to show $\|x_1 - y\| < r_1$ and $\|x_2 - y\| < r_2$. We have by the triangle inequality that

$$\|y - x_1\| \leq \|y - x\| + \|x - x_1\|.$$

We thus need only show this for $\|y - x_1\| = \|y - x\| + \|x - x_1\|$. Note that

$$\|y - x\| < |r_1 - r_2|, \qquad \|x - x_1\| < r_1.$$

We thus have

$$\|y - x_1\| < |r_1 - r_2| + r_1 = |r_2 - r_1 + r_1| = r_1,$$

because $r_1 > 0$. Therefore, $\|y - x_1\| < r_1$. The proof is equivalent for r_2. Therefore, $B_{|r_1-r_2|}(x) \subseteq B_{r_1}(x_1) \cap B_{r_2}(x_2)$, and thus by definition 1.4, $\mathcal{B}$ is a basis for $\mathbb{R}^n$.

Note that the only nontrivial thing used here is the triangle inequality. In the future, we will define a *metric*, which is an abstraction of distance which maintains this, and a few other important properties of distance. We will then find that the collection of all open balls where the norm is replaced by the metric forms a basis for the metric space. This will be called the metric topology.

Note that the basis of a topology is not unique.

Lecture 3: Bases, Closed sets, Hausdorff

Tue 28 Jan 2025 11:05

Proposition 1.2

Let $(X, \mathcal{T})$ be a topological space and let $\mathcal{C}$ be a collection of open subsets of X. Then $\mathcal{C}$ is a basis for the topology $\mathcal{T}$ on X if and only if for all open sets $U \in \mathcal{T}$, for all $x \in U$, there exists $V \in \mathcal{C}$ such that $x \in V \subseteq U$.

Proof. For all $U \in \mathcal{T}$ and for all $x \in U$, let there exist $V \subseteq U$ in $\mathcal{C}$ such that $x \in V$. Since X is open, we have that for all $x \in X$, there exists $V \in \mathcal{C}$ such that $x \in V$. Now let $V_1, V_2 \in \mathcal{C}$. Since $\mathcal{C}$ is a collection of open sets, $V_1 \cap V_2$ is open. We thus have that for all $x \in V_1 \cap V_2$, there exists $V_3 \subseteq V_1 \cap V_2$ in $\mathcal{C}$ such that $x \in V_3$. By definition 1.4, $\mathcal{C}$ is a basis. We now need to show $\mathcal{T}_{\mathcal{C}} = \mathcal{T}$. Let $U \in \mathcal{T}$. For all $x \in U$, there exists $V_x \in \mathcal{C}$ such that $x \in V_x \subseteq U$. We then have

$$U = \bigcup_{x \in U} V_x.$$

$V_x \in \mathcal{C}$, $U \in \mathcal{T}_{\mathcal{C}}$. Additionally, since we defined $\mathcal{C}$ as a collection of *open* sets, and all $V \in \mathcal{T}_{\mathcal{C}}$ are unions of elements in $\mathcal{C}$, by definiton 1.1 we have immediately that $\mathcal{T}_{\mathcal{C}} \subseteq \mathcal{T}$.

Let $\mathcal{C}$ be a basis for $\mathcal{T}$. Let $U \in \mathcal{T} = \mathcal{T}_{\mathcal{C}}$, and let $x \in U$. Since $\mathcal{C}$ is a basis, we can thus write

$$x \in U = \bigcup_{i \in I} V_i = U, \qquad V_i \in \mathcal{C}.$$

Note that x is in this union. Therefore, there exists some $V_i \in \{V_i\}_{i \in I}$ such that $x \in V_i$, and we get for free that

$$V_i \subseteq \bigcup_{i \in I} V_i.$$

Therefore, for all open sets U, for all $x \in U$, there exists $V_i \in \mathcal{C}$ such that $x \in V_i \subseteq U$. ■

Definition 1.6: Closed

A subset $Y \subseteq X$ is *closed* in X if its compliment $Y^c = X \setminus Y$ is open.

For example, in the standard topology on $\mathbb{R}$, a closed interval $[a, b]$ with $a < b$ is closed, because $[a, b]^c = (-\infty, a) \cup (b, \infty)$, and since both of these are open intervals, this set is open.

Lemma 1.1

$$U \in \mathcal{T} \iff \forall x \in U, \exists V_x \in \mathcal{B} \text{ s.t. } x \in V_x \subseteq U.$$

Proof. Follows from previous prp. ■

Using lemma 1.1, we can show the sae for $\mathbb{R}^2$. Let $\overline{B}_\varepsilon(x) = \left\{y \in \mathbb{R}^2 \mid d(y, x) \leq \varepsilon\right\}$. Let $x' \in \overline{B}(x, \varepsilon)^c$, and note that $d(x,x') > \varepsilon$, so $d(x',x) - \varepsilon > 0$. Take

$$= 1/2\gamma = \frac{1}{2}(d(x',x) - \varepsilon) > 0.$$

By definition, $B(x',\gamma)$ is open, and contains x'. It this suffices to show by lemma 1.1 that $B(x',\gamma) \subseteq X \setminus \overline{B}(x,\varepsilon)$. That is, for all $y \in B(x',\gamma)$, $d(x,y) > \varepsilon$. Note that $y \in B(x',\gamma)$ implies

$$d(x',y) < \gamma = \frac{1}{2}\left(d(x',x) - \varepsilon\right) < d(x,x') - \varepsilon.$$

We thus have by the triangle inequality that

$$d(x,y) \geq d(x,x') - d(x',y).$$

We also have

$$d(x,x') - d(x',y) > d(x,x') - \left(d(x,x') - \varepsilon\right) = \varepsilon.$$

Therefore,

$$d(x,y) > \varepsilon,$$

and thus, $B(x',\gamma) \subseteq \overline{B}(x,\varepsilon)^c$. Therefore, for all $x \in \overline{B}(x,\varepsilon)^c$, there exists an open set $x' \in B(x',\gamma) \subseteq \overline{B}(x,\varepsilon)$, and thus $\overline{B}(x,\varepsilon)$ is closed.

Proposition 1.3

Let $(X,\mathcal{T})$ be a topological space. Then

1. $X, \varnothing$ are closed;
2. Finite union of closed sets are closed;
3. Intersections of closed sets are closed.

Proof. Done in the homework. (1) is trivial, and (2,3) follow from De Morgan's laws. ■

Definition 1.7: Hausdorff

A topological space X is Hausdorff if for all distinct $x, y \in X$, there exist open sets U_x, U_y such that $x \in U_x$, $y \in U_y$, and $U_x \cap U_y = \varnothing$.

For example, the trivial topology on $\mathbb{R}$ is *not* Hausdorff. However, we would expect all nice spaces to be Hausdorff, such as $\mathbb{R}$ with the normal topology, so we study Hausdorff spaces in detail.

Here's another example interesting. Let $X = \mathbb{R} \cup \{0'\}$, where $0'$ is some formal point outside of $\mathbb{R}$. Every open set not containing 0 in the standard topology on $\mathbb{R}$ is open, and additionally for all open sets U containing 0, we instead include the new open set U' such that $U' = U \cup \{0'\}$. This is basically just $X = \mathbb{R} \amalg \mathbb{R}/\sim$. This space is *not Hausdorff*, since *every open set* containing 0 must also contain $0'$, so although $0 \neq 0'$, there exist no disjoint neighborhoods of these points. We usually call this a line with double 0.

Lecture 4: I had the flu and missed this class :(

Sun 02 Feb 2025 16:14

Proposition 1.4

If X is Hausdorff, then every singleton $\{x\}$ is closed.

Proof. It suffices to show $\{x\}^c$ is open. Since X is Hausdorff, for all $y \neq x$, there exist disjoint neighborhoods U_x, U_y of x, y, respectively. Let $\{U_\alpha\}_{\alpha \in X \setminus \{x\}}$ be a collection of neighborhoods such that for all α, there exists an open set V_α containing x such that $U_\alpha \cap V_\alpha = \varnothing$. It follows that

$$X \setminus \{x\} = \bigcup_{\alpha \in X \setminus \{x\}} U_\alpha.$$

Therefore, $\{x\}^c$ is closed. ■

Chapter 2

Interior, Closure, and Boundary

We want to be able to talk more precisely about the interior, exterior, and boundary of a topological space.

Definition 2.1: Interior

Let X be a topological space, and let $A \subseteq X$. The *interior* of A, denoted $Int(A)$, is the union of all open sets contained in A. It is the largest open set contained in A.

Definition 2.2: Closure

Let X be a topological space, and let $A \subseteq X$. The *closure* of A, denoted $\overline{A}$, is the intersection of all closed sets containing A. It is the smallest closed set containing A.

Note that by definition, $\text{Int}(A) \subseteq A \subseteq \overline{A}$. This fact will be useful on multiple occasions.

Proposition 2.1

Let X be a topological space, and let $A \subseteq X$. Then

- A is open if and only if $\text{Int}(A) = A$;
- A is closed if and only if $\overline{A} = A$.

Proof. Suppose A is open. Then since A contains itself, and $\text{Int}(A)$ is a union over all open sets contained in A, we have $A \subseteq \text{Int}(A)$. By definition, $\text{Int}(A) \subseteq A$. Now suppose $A = \text{Int}(A)$. Since $\text{Int}(A)$ is a union of open sets, it must be open, and thus A is open.

Suppose A is closed. Then since A contains itself, and $\overline{A}$ is an intersection over all closed sets containing A, we have $\overline{A} \subseteq A$. By definition, $A \subseteq \overline{A}$. Now suppose $A = \overline{A}$. By proposition 1.3, since $\overline{A}$ is an intersection of closed sets, it is closed, and thus A is closed. ■

Here's an interesting example. Consider the finite compliment topology. Closed sets are, by definition, finite, with the sole exception of X. Assuming X is infinite, then $\overline{A} = X$ for all *open* sets A.

Here's another, very important example. Consider $\mathbb{Q} \subseteq \mathbb{R}$ with the standard topology. Note that $\text{Int}(\mathbb{Q}) = \varnothing$, because for any open interval (a, b), there will be an irrational number contained within. Similarly, $\overline{\mathbb{Q}} = \mathbb{R}$.

Definition 2.3: Dense

A subset $A \subseteq X$ of a topological space X is called *dense* if $\overline{A} = X$.

Proposition 2.2

Let $A \subseteq X$, and let X be a topological space. Then

- $\mathrm{Int}(A^c) = \left(\overline{A}\right)^c$;
- $\overline{(A^c)} = \mathrm{Int}(A)^c$.

Proof. The argument is basically the same for both implications, so I will only do the first for convenience. Let $\mathcal{A}$ be the collection of all open sets contained within A, and recall that

$$\mathrm{Int}(A)^c = \left(\bigcup_{U \in \mathcal{A}} U\right)^c.$$

By De Morgan's law, this equals

$$\bigcap_{U \in \mathcal{A}} U^c.$$

Since a set is closed if and only if it's the compliment of an open set, this collects all closed sets containing A^c. Therefore, this is $\left(\overline{A}\right)^c$. ■

Proposition 2.3

Let X be a topological space, and let $A \subseteq X$. Then $x \in \mathrm{Int}(A)$ if and only if there exists an open neighborhood $U \subseteq A$ of x. Additionally, $y \in \overline{A}$ if and only if every neighborhood of y intersects A.

Proof. The first proposition is trivial. $\mathrm{Int}(A)$ is open by definition, so an open neighborhood exists. If an open neighborhood $U \subseteq A$ exists, then $x \in \mathrm{Int}(A)$ by definition.

For the second proposition, suppose $y \in \overline{A}$, and suppose for contradiction there exists some open set U containing y such that $U \cap A = \varnothing$. It follows that $U \subseteq A^c$, and thus $y \in \mathrm{Int}(A^c)$ by the first proposition. By proposition 2.2, $\mathrm{Int}(A^c)^c = \overline{A}$, and thus we have $y \notin \overline{A}$, a contradiction.

Now suppose for all neighborhoods U of y that $U \cap A$ is nonempty. Suppose for contradiction that $y \notin \overline{A}$. It follows that $y \in \overline{A}^c$, and by proposition 2.2, $y \in \mathrm{Int}(A^c)$. By the first proposition, there exists an open set $U \subseteq A^c$, and thus $U \cap A = \varnothing$, a contradiction. ■

This proposition motivates the following definition.

Lecture 5: Limit Points

Tue 04 Feb 2025 10:55

Definition 2.4: Limit Point

Let X be a topological space, and let $A \subseteq X$. A point x is a *limit point* of A if every open set U containing x intersects A at a point other than x.

Proposition 2.4

Let A' denote the set of limit points of A. Then $\overline{A} = A \cup A'$. Equivalenty, $\overline{A} \setminus A = A'$.

Proof. Recall by proposition 2.3 that $x \in \overline{A}$ if and only if every open neighborhood of x intersects A. Note that if $x \in \overline{A} \setminus A$, then $x \notin A$, but since $x \in \overline{A}$, we still have that every open set must intersect A. Therefore, every open set containing A will intersect A at some point other than x. By definition 2.4, x is a limit point of A. Since proposition 2.3 was an if and only if, we have the equality rather than a subset. ∎

This proposition motivates a definition of convergence for sequences in topological spaces.

Definition 2.5: Convergence

Let X be a topological space, and let $\{x_n\}$ be a sequence in X. We say

$$\lim_{n\to\infty} x_n = x$$

if for any open set U containing x, there exists $N \in \mathbb{N}$ such that $x_n \in U$ for all $n \geq N$. We then say the sequence *converges to* x, and write $x_n \to x$.

For example, in the standard topology on $\mathbb{R}$, if we let $x_n = 1/n$, then clearly

$$\lim_{n\to\infty} x_n = 0.$$

We also have that for any open set U containing 0, there exists $a < 0 < b$ such that $(a, b) \subseteq U$. By the Archimedean property, there exists N such that $1/n < b$ if $n > N$.

Note that the limit of the sequence may not be unique. For example, consider the trivial topology on $\mathbb{R}$, and once again consider $x_n = 1/n$. We then have

$$\lim_{n\to\infty} x_n = 2025.$$

Since the only open set containing points is $\mathbb{R}$, every point is a limit point. The trivial topology has too few open sets to properly separate points.

Proposition 2.5

Let X be Hausdorff. Then any convergent sequence has a unique limit point.

Proof. Let $\{x_n\}$ converge to x. Suppose for contradiction that there exists another distinct limit point y of $\{x_n\}$. There then exist disjoint open sets U, V containing x, y, respectively. By definition, there then exists $N \in \mathbb{N}$ such that $x_n \in U$ for all $n \geq N$. However, $x_n \notin V$, because $U \cap V = \varnothing$. Therefore, there exists an open set V of y such that $x_n \notin V$ for all $n \geq N$, so $\{x_n\}$ cannot converge to y, a contradiction. ∎

Definition 2.6: Boundary

Let X be a topological space, and let $A \subseteq X$. Define

$$\partial A := \overline{A} \setminus \operatorname{Int}(A).$$

We call ∂A the *boundary* of A.

Proposition 2.6

Let X be a topological space, and let $A \subseteq X$. Then $x \in \partial A$ if and only if for all neighborhoods U of x such that $U \cap A \neq \varnothing$, we have $U \cap A^c \neq \varnothing$.

Proof.

$$\begin{aligned} x \in \partial A &\iff x \in \overline{A} \setminus \operatorname{Int}(A) \\ &\iff x \in \overline{A} \wedge x \notin \operatorname{Int}(A) \\ &\iff x \in \overline{A} \wedge x \in \operatorname{Int}(A)^c = \overline{A^c}. \end{aligned}$$

Therefore, for every open set U containing x, we have $x \in U \cap \overline{A}, U \cap \overline{A^c}$, and thus the statement is proven. ■

It's very natural at this point to ask if we can construct more topological spaces, given a selection of spaces we aready know a lot about.

Definition 2.7: Subspace Topology

Let $(X, \mathcal{T})$ be a topological space, and let $A \subseteq X$. Then define

$$\mathcal{T}_A \coloneqq \{U \cap A \mid U \in \mathcal{T}\}.$$

We say this is the subspace topology on A.

The claim is clearly that this is a topology on A. We will show this using the definition. Clearly, $\varnothing, A \in \mathcal{T}_A$.

Now let $\{U_1, \ldots, U_n\} \subseteq \mathcal{T}_A$. We can write

$$U_i = U_i' \cap A$$

for all i. We then have

$$\bigcap_{i=1}^{n} U_i = \bigcap_{i=1}^{n} U_i' \cap A = \left(\bigcap_{i=1}^{n} U_i'\right) \cap A.$$

Therefore,

$$\bigcap_{i=1}^{n} U_i \in \mathcal{T}_A.$$

Finally, let $\{U_i\}_{i \in I} \subseteq \mathcal{T}_A$. Let

$$U_i = U_i' \cap A$$

for all i. We have

$$\bigcup_{i \in I} U_i = \bigcup_{i \in I} (U_i' \cap A) = \left(\bigcup_{i \in I} U_i'\right) \cap A.$$

Therefore,

$$\bigcup_{i \in I} U_i \in \mathcal{T}_A.$$

We thus have that $(A, \mathcal{T}_A)$ is a topological space.

Proposition 2.7

Let X be a topological space, and let $A \subseteq X$ be a nonempty set equipped with the subsapce topology. Then $D \subseteq A$ is closed if and only if $D = C \cap A$ for some closed set $C \subseteq X$.

Proof. By definition, we must have the compliment of D in A is open. Define $U \coloneqq A \setminus D$. Note that since U is open, there must exist an open set $U' \subseteq X$ such that $U = U' \cap A$. It follows that $A \setminus D = U' \cap A$. Something happens I don't know and I don't think the professor does either LMAO ■

Proposition 2.8

Let X be a topological space, and let $A \subseteq X$ be equipped with the subspace topology. If $\mathcal{B}$ is a basis for the topology on X, then

$$\mathcal{B}_A \coloneqq \{B \cap A \mid B \in \mathcal{B}\}$$

is a basis for the subspace topology on A.

Proof. yeah i REALLY dont wanna do this. ■

Lecture 6: Product and Quotient Topologies

Thu 06 Feb 2025 11:02

A classical example is the unit circle $S^1 \subseteq \mathbb{R}^2$, where $\mathbb{R}^2$ is given the standard topology. The unit circle can then be given the subspace topology, generated by open arcs along the circle, since every intersection of an open ball intersecting the circle will yield an open arc. We call this the standard topology on S^1. Some people may also use the notation S^1_R for the 1-sphere with radius R.

We are still looking for ways to create new topologies out of old topologies. For example, the product topology.

Definition 2.8: Product Topology

Let X, Y be topological spaces with bases $\mathcal{B}_X, \mathcal{B}_Y$. Define

$$\mathcal{B}_{X \times Y} \coloneqq \{U \times V \mid U \in \mathcal{B}_X, V \in \mathcal{B}_Y\}.$$

We call the topology generated by this set the *product topology* on $X \times Y$.

The product space comes with natural projections $p_1 : X \times Y \to X$, $p_2 : X \times Y \to Y$ given by $(x, y) \mapsto x$ and $(x, y) \mapsto y$, respectively.

For example, consider $S_1 \times S_1$. We have two projections

$$\begin{tikzcd} S_1 \times S_1 \arrow[r] \arrow[d] & S_1 \\ S_1 & \end{tikzcd}$$

Imagine placing a circle S^1 at each point, going along another circle. You get a taurus.

Similarly, consider $S^1 \times \mathbb{R}^+$. This can be identified with a cylinder (stringing circles along the real line), or it can also be identified as a set with $\mathbb{R}^2$, by using polar coordinates.

Now we will do a more indepth example. Consider $\mathbb{R} \times \mathbb{R}$ with the standard topology. Each basis element is thus the product of open intervals, so a rectangle. We showed in the first homework the collection of open rectangles generate the standard basis on $\mathbb{R}^2$.

Now for a more formal tratment of the 2-torus. The standard topology on the Torus looks like it's generated from open rectangles from it's surface. The claim is that $S^1 \times S^1$ gives this standard topology (makes sense), so we need to show this is really a basis. We can actually just prove this in a more general sense.

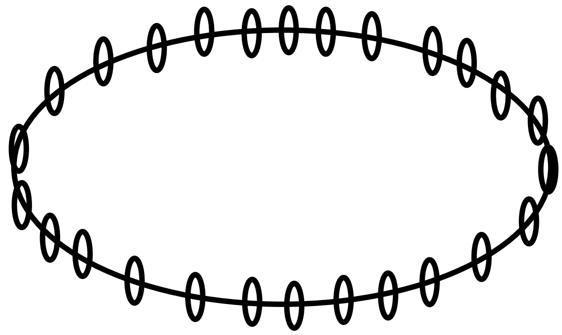

Figure 2.1: $S^1 \times S^1$

Theorem 2.1

The basis $\mathcal{B}_{X\times Y}$ is a basis, and is independent of choice of $\mathcal{B}_X, \mathcal{B}_y$.

Proof. For all $(x, y) \in X \times Y$, there exists $U \times V \in \mathcal{B}_{X\times Y}$ containing (x, y) since $\mathcal{B}_X, \mathcal{B}_Y$ are bases, Additionally, let $U_1 \times V_1, U_2 \times V_2 \in \mathcal{B}_{X\times Y}$, and let (x, y) be contained within

$$(U_1 \times V_1) \cap (U_2 \times V_2).$$

Since $\mathcal{B}_X, \mathcal{B}_Y$ are bases, there exist $U_3 \subseteq U_1 \cap U_2$ and $V_3 \subseteq V_1 \cap V_2$ satisfying this stuff, so we have that

$$(x, y) \in U_3 \times V_3 \subseteq (U_1 \times V_1) \cap (U_2 \times V_2).$$

Therefore, this is a basis. Finally, we must show the choice of basis is irrelevant. Let $\mathcal{B}_X, \mathcal{B}'_X$ be bases X, and let $\mathcal{B}_Y, \mathcal{B}'_Y$ be bases for Y. It suffices to show that

$$\mathcal{T}_{\mathcal{B}_{X\times Y}} \subseteq \mathcal{T}_{\mathcal{B}'_{X\times Y}}.$$

This is because we can use the same statement in the other direction to show equality.

Let

$$\bigcup_{i\in I} U_i \times V_i \in \mathcal{B}_{X\times Y},$$

where $U_i \in \mathcal{B}_X$, $V_i \in \mathcal{B}_Y$ for all $i \in I$. Fix $x \in U_i$. By properties of basis, there exists $U'_{i,x} \in \mathcal{B}'_X$ such that

$$x \in U'_{i,x} \subseteq U_i.$$

We can thus write
$$U_i = \bigcup_{x \in U_i} U'_{i,x}.$$
We can apply the same logic to the V_is, showing that it can be written as a union of elements in $\mathcal{B}'_Y$. Therefore, we are done. ■

We can create yet another topology, called the quotient topology.

Definition 2.9: Quotient Topology

Let A be a nonempty set and let X be a topological space. Let $p : X \twoheadrightarrow A$ be surjective. Define the quotient topology on A such that $U \subseteq A$ if open if and only if $p^{-1}(U)$ is open in X.

Proposition 2.9

Let A be a nonempty set and X a topological space, and let $p : X \twoheadrightarrow A$ a surjection. The quotient topology on A by p defines a topology on A.

Proof. Let $\mathcal{T}_A$ be the collection of all subsets of A with open preimages. We first want to show $\varnothing, A \in \mathcal{T}_A$. Since p is surjective, $p^{-1}(\varnothing) = \varnothing$. Additionally, since p is surjective, $p^{-1}(A) = X$. Since $\varnothing, X$ are open in X, we have $\varnothing, A \in \mathcal{T}_A$ are open.

Now let $U_1, \ldots, U_k$ be open in A. Note that since $p^{-1}(U_i)$ is open in X, and finite intersection of open sets is open,
$$\bigcap_{i=1}^{k} p^{-1}(U_i)$$
is open in X. Therefore,
$$\bigcap_{i=1}^{k} U_i \in \mathcal{T}_A.$$
Finally, let $\{U_i\}_{i \in I} \subseteq \mathcal{T}_A$. Then by the same logic,
$$\bigcup_{i \in I} p^{-1}(U_i)$$
is open in X. It follows that
$$p\left(\bigcup_{i \in I} p^{-1}(U_i)\right) = \bigcup_{i \in I} U_i \in \mathcal{T}_A.$$
■

For example, consider $\mathbb{R}$ and S^1, with the map $\mathbb{R} \to S^1$ given by $t \mapsto \exp(2\pi i t)$. The prof is lost trying to explain what this topology is. We can identify $[0,1)$ with S^1 by the map p. Ok so an open set is basically the image of an interval, which is going to be an open arc, so the quotient topology is the standard topology on S^1.

Given any relation $\sim$ on X, consider the quotient map in **Set** given by $x \mapsto [x]$. Given that $(X, \mathcal{T})$ is a topological space, we can then give $X/{\sim}$ the quotient topology by the quotient map. An application of this is taking the relation $(0, y) \sim (1, y)$ on $[0,1] \times [0,1]$. This glues together two sides of the rectangle, creating the cylinder $S^1 \times [0,1]$.

Lecture 7: Quotient Topologies

Tue 11 Feb 2025 11:08

A quotient can be redefined in the more intuitive way. Consider a map $f : A \to X$ with $A \subseteq X$. We have the diagram

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[dr, "\pi"] & X \\ & f(A) \arrow[u] \end{tikzcd}$$

We can define the relation $\sim$ on X such that $x \sim y$ if and only if $y = f(x)$. This is trivially an equivalence relation, so we then have a surjection $X \to X/{\sim}$ given by the canonical projection, and thus any function $f : A \to X$ gives rise to a very natural quotient topology.

IDK what's happening. Consider $X = [0, 1]$ with the subspace topology, and let $A = \{0\}$. Consider the quotient map $f : A \to X$ given by $0 \mapsto 1$. Then the quotient topology [check the notes i see a diagram this looks interesting]

Consider $X = [0, 1] \times [0, 1]$. This is just the closed square in $\mathbb{R}^2$. Let $A = \partial X$, and consider the map $A \to X$ by $(0, t) \mapsto (1, t)$, $(t, 0) \mapsto (t, 1)$, and otherwise the identity map. We saw already gluing together two sides gives a cylinder, but now we are gluing together the top and bottom as well. This can be seen as gluing together the circular sides of the cylinder, giving a torus.

Consider the same spaces, but a new map this time. We are mapping $\{0\} \times [0, 1] \to \{1\} \times [0, 1]$ and $[0, 1] \times \{0\} \to [0, 1] \times \{1\}$, but this time we define it as $(0, t) \mapsto (1, t)$ and $(t, 0) \mapsto (1 - t, 1)$. Let's ignore this first map and only consider the mapping in y. We are gluing together the top side once again, but flipping the direction. This gives us a Mobius strip, which only has one side, and is thus non-orientable.

Now let's return to considering both the x and y mappings, We are gluing together 2 ends of a cylinder, but gluing together their ends in two opposite directions. We can't do this in 3d, but we can do it formally in mathematics. What we get is a Klein bottle. It can be shown that the Klein bottle cannot be embedded in $\mathbb{R}^3$ in a *proper* way, such that no self-intersections happen.

Here's another non-orientable example. Consider the unit disc $X = \left\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\right\}$ with the subspace topology, and consider $A \coloneqq \partial X = S^1$. Consider the map $A \to X$ by $(x, y) \mapsto (-x, -y)$. We call the corresponding space $X/{\sim}$ the real projective space $\mathbb{RP}^2$. As a set, this is simply the set of lines in $\mathbb{R}^2$ passing through the origin. Also note that

$$\mathbb{RP}^2 \cong \mathbb{R}^3 \setminus \{0, 0, 0\} /{\sim},$$

where $\sim$ is defined as

$$[(x, y, z)] = \{\lambda(x, y, z) \mid \lambda \in \mathbb{R}\}.$$

This is kind of because you can use a bijection to identify the unit disc D^2 with the entirety of $\mathbb{R}^2$. The real projective space is non-orientable, because it's kind of like once you get to the boundary of the disc ∂X, the map sends you to the other side, and you have to walk across it again.

Consider two surfaces S_1, S_2. We can construct the connected sum of S_1 and S_2, denoted $S_1 \# S_2$. Choose a point $p \in S_1$, and delete a sphere at p. So if p is a 2d surface in $\mathbb{R}^3$, the sphere would be a 2-sphere, which is a disk. Now choose a point $g \in S_2$, and do the same. Call the corresponding spheres S_p and S_g. The boundaries are simply the unit sphere of one lower dimension, so we can map them together. We choose a map $f : \partial S_p \to \partial S_g$, and this gives an equivalence relation. We then have that

$$S_1 \# S_2 = S_1 \amalg S_2 / \sim_f.$$

This basically glues two orientable surfaces (manifolds) together. Note that orientable surfaces are homeomorphic if and only if they have the same genus (defined as number of "holes", we will define later). This means the connected sum really just adds together the genuses.

Definition 2.10: Connected Sum

Let S_1, S_2 be n-manifolds. Choose points $p \in S_1$, $q \in S_2$, and remove an n-sphere at each point. Given any map f between the boundaries of these n-spheres, define the connected sum $S_1 \# S_2$ as

$$S_1 \# S_2 \coloneqq S_1 \amalg S_2 / \sim_f .$$

One final example. Let $f : M \to M$ be a bijection for M a topological space. We define the mapping torus M_f as $M \times [0,1]/\sim$, where

$$(x, 0) \sim (f(x), 1).$$

For example, if $f : S^1 \to S^1$ is the identity map, then $S^1_f \cong T^2$ is simply the torus.

Lecture 8: Continuity and Homeomorpisms

Thu 13 Feb 2025 10:59

Definition 2.11: Continuous

Let X, Y be topological spaces, and let $f : X \to Y$. We say f is continuous if and only if for all open sets $U \subseteq Y$, the preimage $f^{-1}(U)$ is open.

For example, $\mathrm{id}_X : X \to X$ is clearly continuous. Also, any constant map is continuous. Let $f : X \to Y$ be the map $x \mapsto y$ for a fixed $y \in Y$. Then for any open set containing y, the preimage is X, and for any open set not containing y, the preimage is $\varnothing$.

Another example is the quotient topology $q : X \twoheadrightarrow A$. Note that A with the quotient topology is defined such that for all open sets U in A, the preimage $q^{-1}(U)$ is open in X. Therefore, the quotient map q is continuous by definition. Also consider the product topology $X \times Y$ with the projections π_X, π_Y. These are clearly continuous. GIven an open set $V \subseteq X$, the preimage is $V \times Y$, and since V is open in X and Y is open in Y, by definition 2.8, this is open. Finally, Let $A \subseteq X$ and consider the inclusion $A \hookrightarrow X$, where A has the subspace topology. Note that $\iota^{-1}(V) = A \cap V$, so the inclusion is continuous.

Proposition 2.10

Let X, Y be topological spaces, and let $\mathcal{B}$ be a basis for the topology on Y. Then $f : X \to Y$ is continuous if and only if $f^{-1}(B)$ is open for all $B \in \mathcal{B}$.

Proof. If f is continuous, then this follows by definition 2.11.

Let $f^{-1}(B)$ be open in X for all $B \in \mathcal{B}$. Let $V \subseteq Y$ be open. It follows that there exists a collection $\{B_i\}_{i \in I} \subseteq \mathcal{B}$ such that

$$f^{-1}(V) = f^{-1}\left(\bigcup_{i \in I} B_i\right) = \bigcup_{i \in I} f^{-1}(B_i).$$

This is a union of open sets, so $f^{-1}(V)$ is open in X. ■

Fr example, $f : \mathbb{R} \to \mathbb{R}$ with the standard topology is continuous if and only if $f^{-1}(B_\delta(y))$ is open in $\mathbb{R}$, for all $\delta > 0$, and for all $y \in \mathbb{R}$. Note that we can write this in the basis for the first $\mathbb{R}$:

$$f^{-1}(B_\delta(y)) = \bigcup_{i \in I} B_{\delta_i}(x_i)$$

for some collection of $\delta_i > 0$, $x_i \in \mathbb{R}$. If $a \in \mathbb{R}$ such that $f(a) = b$, then $a \in f^{-1}(b)$, and

$$b \in f^{-1}(B_\delta(b)) = \bigcup_{i \in I} B_{\delta_i}(x_i).$$

Note that a is inside of this union, since $f^{-1}(b)$ is a subset of it, and thus $B_\varepsilon(a) \subseteq B_{\delta_i}(x_i)$ for all i for $\varepsilon = \min\{|a-(x_i-\delta_i)|, |a-(x_i+\delta_i)|\}$. We have

$$B_\varepsilon(a) \subseteq B_{\delta_i}(x_i) \subseteq \bigcup_{i\in I} B_{\delta_i}(x_i) = f^{-1}(B_\delta(b)).$$

Therefore, for any $\delta > 0$, there exists an $\varepsilon > 0$ such that $|x-a| < \delta$ implies that $|f(x)-f(a)| < \varepsilon$. That is, $f : \mathbb{R} \to \mathbb{R}$ is continuous if

$$\forall\delta > 0 \exists\varepsilon > 0 : |x-a| < \delta \implies |f(x)-f(a)| < \varepsilon.$$

Theorem 2.2

A function $f : X \to Y$ is continuous if and only if for all $x \in X$ and $U \subseteq Y$ open such that $f(x) \in U$, there exists an open neighborhood $V \subseteq X$ of x such that $f(V) \subseteq U$.

Proof. Let f be continuous, and $x \in X$, and let $f(x) \in U \subseteq Y$ be open. Then the preimage $f^{-1}(U)$ contains x. We now want to find a neighborhood V of x such that $f(V) \subseteq U$. We can take $V = f^{-1}(U)$. Since f is continuous, V is open, and clearly $f(V) = U \subseteq U$.

Now for the converse. Suppose for any $x \in X$ and $f(x) \in U$ open in Y, then there exists a neighborhood $V \subseteq X$ of x such that $f(V) \subseteq U$. We want to show $f^{-1}(U)$ is open. For every $x \in X$ such that $f(x) \in U$, we have that there exists an open set $x \in V_x$ such that $f(V_x) \subseteq U$. It follows that

$$f^{-1}(U) = \bigcup_{x \in X \text{ s.t. } f(x) \in U} V_x.$$

This is a union of open sets by assumption. I thought this previous line was the most trivial claim we have made today, but apparently it's the only one that needs explaining. Sure. ■

Theorem 2.3

Let $f : X \to Y$ be continuous, and let $x_n \to x$ be a convergent sequence in X. Then $f(x_n) \to f(x)$ in Y.

Proof. Recall the definition of a limit of a sequence (2.5) let $\{x_n\} \subseteq X$. Then

$$\lim_{x\to\infty} x_n = x$$

if and only if for any open set U containing x, there exists $N \in \mathbb{N}$ such that $x_n \in U$ for all $n \geq N$. Now let $x_n \to x$. We wish to show for all open $V \in Y$ containing $f(x)$, there exists $N \in \mathbb{N}$ such that $f(x_n) \in V$ for all $n \geq N$. Since V is open, we have that $f^{-1}(V)$ is open. Note that for all $f(x_n) \in V$, we have $x_n \in f^{-1}(V)$. There thus exists $N \in \mathbb{N}$ such that $x_n \in f^{-1}(V)$ if $n \geq N$. Therefore, there exists $N \in \mathbb{N}$ such that $f(x_n) \in V$ if $n \geq N$, and $f(x_n) \to f(x)$. ■

Consider the example $f : \mathbb{R} \to \mathbb{R}$ given by $x \mapsto x^2$. Suppose $b \in f(\mathbb{R})$, given $\varepsilon > 0$, we want to find $\delta > 0$ such that $|x-a| < \delta$ implies $|f(x)-f(a)| < \varepsilon$. Note that

$$|f(x)-f(a)| = \left|x^2-a^2\right| = |x-a|\,|x+a| < \delta\,|x-a|.$$

Choose $\delta = \varepsilon / |x+a|$. We have that

$$|f(x)-f(a)| < \frac{\varepsilon}{|x+a|}\,|x+a| = \varepsilon.$$

There is some manipulation to turn $|x+a|$ into $|3a|$ using some fact that I didn't see. But using this assumption f is continuous. Well if $a=0$ then it breaks so we choose 1 or something.

Proposition 2.11

Let $f : X \to Y$, $g : Y \to Z$ be continuous. Then $gf : X \to Z$ is continuous.

Proof. Let $U \subseteq Z$ be open. Since g is continuous, $g^{-1}(U)$ is open in Y. Since f is continuous, $f^{-1}(g^{-1}(U))$ is open in X. Note that

$$(gf)^{-1}(U) = f^{-1}(g^{-1}(U)).$$

■

Using this fact, and the fact that the identity map 1_X is continuous, we have that topological spaces form a category **Top** where the morphisms are continuous maps. We can now define isomorphisms in this category.

Definition 2.12: Homeomorphism

A function $f : X \to Y$ is a homeomorphism if and only if

- f is bijective (the inverse is well-defined);
- f is continuous;
- and f^{-1} is continuous.

If there exists a homeomorphism $f : X \to Y$, we say X is homeomorphic to Y, and we write $X \cong Y$.

Very clearly, these are isomorphisms in **Top**. For example, consider $(-1,1) \cong \mathbb{R}$ with the standard topology. Consider the map

$$x \mapsto \frac{x}{1+|x|}.$$

It's easy to see that $f(\mathbb{R}) \subseteq (-1,1)$. Also note that f is strictly increasing, and thus it is bijective between its limits, which are -1 and 1. Alternatively, the map $1+|x| : (-1,1) \to \mathbb{R}$ is continuous, and is clearly an inverse. Finally, we can show that f is continuous by showing that $1/f^{-1}$ is continuous, and using proposition 2.11 together with the identity map. Therefore, $(-1,1) \cong \mathbb{R}$.

Another example is $[0,1) \to S^1$ given by the map $x \mapsto (\cos(2\pi x), \sin(2\pi x))$. f is trivially bijective, and is continuous because sin and cos are continuous, so their product will be continuous. HOWEVER, f^{-1} is not continuous. Note that $[0,\delta)$ is open in $[0,1)$, but this will map to a partially closed interval in S^1. This is not open, so f^{-1} is not continuous. Therefore, f is not a homeomorphism.

Chapter 3

Metric Spaces

Lecture 9: Metric Spaces

Thu 20 Feb 2025 10:54

Definition 3.1: Metric

Let X be a set, and let $d : X \times X \to \mathbb{R}$. Then d is called a *metric* if

- $d(x, y) = 0$ if and only if $x = y$;
- $d(x, y) = d(y, x)$;
- (Triangle inequality) $d(x, y) + d(y, z) \geq d(x, z)$.

We say the set X together with the metric d, denoted (X, d), is a metric space.

Moreover, if we only have $d(x, x) = 0$ (one-direction implication instead of iff), then we call d a pseudo metric. If (X, d) is a pseudo-metric space, we can define an equivalence relation $x \sim y$ if and only if $d(x, y) = 0$. It then follows fairly naturally that $(X/{\sim}, d)$ is a metric space.

Examples of metric spaces are very abundant. For example, $\mathbb{R}^n$ together with the usual distance function $d(x, y) = \|x - y\|$. An example of a pseudo-metric space is the space of functions on $[0, 1]$, with the metric defined by

$$d(f, g) = |f(0.78) - g(0.78)| .$$

This works for any number in $[0, 1]$, not just 0.78. Clearly $f(0.78) = g(0.78)$ does not imply that $f = g$, but clearly all other claims hold.

Definition 3.2: Metric Topology

Let (X, d) be a metric space, and let $B_r(x)$ be the open ball of radius r about x, defined as

$$B_r(x) = \{y \in X \mid d(x, y) < r\} .$$

Then the collection of all such open balls $\mathcal{B}$, given as

$$\mathcal{B} = \left\{B_r(x) \mid r \in \mathbb{R}^+, x \in X\right\}$$

forms a basis for the *metric topology* induced by d on X.

Showing that this is indeed a basis is simple, but messy. One important class of metric spaces is the L^p

spaces. Given continuous functions f, g on $[a, b]$, the L^p metric is defined as

$$\|f-g\|_p = \left(\int_a^b |f-g|^p \, \mathrm{d}x\right)^{1/p}.$$

Note that the L^p spaces are more general than just continuous functions, but the norm applies to continuous functions as well.

Theorem 3.1

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \to Y$. Then f is continuous if and only if for all $\varepsilon > 0$, for all $x \in X$, there exists $\delta > 0$ such that $f(B_\delta(x)) \subseteq B_\varepsilon(f(x))$.

Proof. Let f be continuous. Note that $f(x) \in B_\varepsilon(f(x))$ clearly, and thus $f^{-1}(B_\varepsilon(f(x)))$ is open in X, since f is continuous. We have x in the preimage fairly trivially. Since this forms a basis, there exists an element of the basis $B_{\delta'}(y)$ containing x such that $B_\delta(y) \subseteq f^{-1}(B_\varepsilon(f(x)))$. Therefore, $d(x, y) < \delta'$. Take $\delta = \delta' - d(x, y)$. It follows from some messy juggling of the triangle inequality that $B_\delta(x) \subseteq B_{\delta'}(y) \subseteq f^{-1}(B_\varepsilon(x))$. We can apply f on both sides to obtain $f(B_\delta(x)) \subseteq B_\varepsilon(x)$.

Now for the converse. Let $U \subseteq Y$ be open. We wish to show $f^{-1}(U)$ is open in X. Let $x \in f^{-1}(U)$, and thus $f(x) \in U$. By propert of basis, there exists $B_{\varepsilon'}(y)$ such that $f(x) \in B_\varepsilon(y) \subseteq U$. Now consider the ball of radius $\varepsilon = \varepsilon' - d(y, f(x))$ centered at $f(x)$. Once again,

$$B_\varepsilon(f(x)) \subseteq B_{\varepsilon'}(y) \subseteq U.$$

By assumption, we have that there exists $\delta > 0$ such that $f(B_\delta(x)) \subseteq B_\varepsilon(f(x))$. Equivalently, $B_\delta(x) \subseteq f^{-1}(B_\varepsilon(f(x)))$. This means for all $x \in f^{-1}(U)$, there exists a basis element that is a subset of the preimage containing x. Taking the union over all x in the preimage gives us that $f^{-1}(U)$ is open. ■

Another example of a metric space is given by the Hausdorff distance. Let (X, d) be a metric space with $A, B \subseteq X$. The question is whether we can measure the distance between two *subsets* of X. Define

$$d_H(A, B) = \inf \{\varepsilon \mid A \subseteq B_\varepsilon(B), B \subseteq B_\varepsilon(A)\}.$$

In this case, the notation $B_\varepsilon(A)$ means

$$B_\varepsilon(A) = \{x \in X \mid \exists a \in A \text{ s.t. } d(x, a) < \varepsilon\}.$$

This can be thought of intuitively, albeit not precisely, as the distance from the edge of A. This is an ε-neighborhood of A.

Lecture 10: More on Metric Spaces

Tue 25 Feb 2025 11:04

Given a collection $\{(X_i, d_i)\}_{i \in I}$ of metric spaces, we can define the product

$$X \coloneqq \prod_{i \in I} X_i.$$

This space is a metric space under the metric

$$d = \sum_{i \in I} d_i.$$

Suppose now that I is contable, and let $x = (x_1, x_2, \ldots)$ and y similarly defined by elements of X. Then

$$d(x,y) = \sum_{i=1}^{\infty} \frac{d_i(x_i, y_i)}{2^i\left(1 + d_i(x_i, y_i)\right)}.$$

We add the extra factor to ensure the sequence converges.

Definition 3.3: Standard Bounded Metric

Let (X, d) be a metric space. Then the *standard bounded metric* $\overline{d}$ is the function

$$\overline{d}(x,y) = \min\left\{1, d(x,y)\right\}.$$

This is a metric on X fairly trivially.

Here's another example. Let (X, d) be a metric space with $A \subseteq X$. Then there's a natural metric space given by the restriction $(A, d|_{A\times A})$.

We know that a metric induces a topology. However, is the converse true? Given a topologial space $(X, \mathcal{T})$, is there an induced metric d on X such that the induced topology on X is $\mathcal{T}$? In other terminology, is X metrizable? There is a famous theorem that somewhat answers this. THe lemma is actually more important somehow.

Definition 3.4: Normal

A topological space X is normal if and only if for any two disjoint closed sets A, B, there exist disjoint neighborhoods U, V of A and B.

Lemma 3.1 (Urysohn)

Let $(X, \mathcal{T})$ be a topological space. Then X is normal if and only if any two disjoint closed sets can be separated by continuous functions. This means there exists a continuous function $f : X \to \mathbb{R}$ such that $f(A) \subseteq \{0\}$ and $f(B) \subseteq \{1\}$.

Proof. Start by supposing all closed A and B can be separated by continuous functions. Note that $U = f^{-1}(1/2, \infty)$ is open, and $V = f^{-1}(-\infty, 1/2)$ is also open. These intervals are disjoint, and we have $A \subseteq U$ and $B \subseteq V$ fairly easily. Therefore, X is normal.

Now suppose X is normal, and let A, B be disjoint closed sets. Let $r = \frac{a}{2^n}$ for some $n \in \mathbb{N}$, and $1 \leq a \leq 2^{n-1}$. We claim there exists an open set $U(r) \subseteq X$ and closed set $V(r) \subseteq X$ such that $A \subseteq U(r) \subseteq V(r) \subseteq B^c$, and that $V(r) \subseteq U(s)$ if $r < s$. We also claim that we can define $f : X \to \mathbb{R}$ such that

$$x \mapsto \inf\left\{r = \frac{a}{2^n} \mid x \in U(r), a, n \in \mathbb{N}\right\}.$$

Moreover, we claim f is continuous and bounded below by 0. I zoned out during this proof becuase it's going so slow and involves numbers. ■

Theorem 3.2 (Urysohn Metrization Theorem)

Let $(X, \mathcal{T})$ be a topological space such that

- Singletons are closed;
- For all $x \in X$ and $Y \subseteq X$ closed with $x \notin Y$, then there exist disjoint neighborhoods of x and Y;
- The topology admits a coubtable basis.

Then $(X, \mathcal{T})$ is metrizable.

Remark

It in fact follows that if singletons are closed and points and closed sets have disjoint neighborhoods (1, 2 in last theorem) then the space is normal.

Lecture 11: Connected

Thu 06 Mar 2025 11:05

Lemma 3.2

Let X be a topological space. If $\forall x \in X$,

- $\{x\}$ is open;
- For all $Y \subseteq X$ with $x \notin Y$, there exist disjoint open sets U, V such that $x \in U$, $Y \subseteq V$;
- X has a countable basis.

THen X is *normal*, meaning for any two closed disjoint subsets $A, B \subseteq X$, there exist disjoint open neighborhoods U, V with $A \subseteq U$, $B \subseteq V$.

Proof. Let $A, B \subseteq X$ be closed. Fix $x \in A$. By assumption, there exist U_x, V disjoint open sets containing x and B. Now consider a collection $\{U_x\}_{x \in A}$, and note that

$$A \subseteq \bigcup_{x \in A} U_x \subseteq B^c.$$

This gives us an open neighborhood of A. Moreover, there exists a countable basis $\{B_i\}_{i \in \mathbb{N}}$. We can replace each U_x with a collection of basis elements, meaning there exists $I \subseteq \mathbb{N}$ and a collection $\{B_i\}_{i \in I}$ such that the union above is equal to

$$A \subseteq \bigcup_{i \in I} B_i.$$

We can do the same procedure; that is, there exists another collection $\{B_j\}_{j \in J}$ with

$$B \subseteq \bigcup_{j \in J} B_j \subseteq A^c.$$

We now have two neighborhoods of A and B respectively. Let $\widetilde{U}$ be defined as the union

$$\widetilde{U} \coloneqq \bigcup_{n \in \mathbb{N}} \left(B_n \setminus \bigcup_{j \in J} \overline{B}_j \right).$$

Similarly, defined

$$\widetilde{V} \coloneqq \bigcup_{n \in \mathbb{N}} \left(B_n \setminus \bigcup_{i \in I} \overline{B}_i \right).$$

In this case, B_n ranges over all basis elements. These are open, because setminus is an intersection with the compliment, which is open in this case, since this is a finite union of closed sets. We still have $A \subseteq \widetilde{U}$, since we are removing everything in the compliment of A. Similarly, $B \subseteq \widetilde{V}$. These sets are fairly obviously disjoint, so this concludes the proof of the Metrization theroem. ■

Chapter 4

Connectedness and Compactness

Definition 4.1: Connected

Let X be a topological space. We say X is *connected* if it does not have disjoint proper nonempty open sets $U, V \subsetneq X$ such that $U \cup V = X$.

If X is not connected, we say X is *disconnected*. We say a subset $A \subseteq X$ is (dis)connected if it is (dis)connected with respect to the subspace topology.

For example, $\mathbb{R}$ with the finite compliment topology is obviously connected, since there exist no disjoint nonempty open sets.

Proposition 4.1

Let X be connected, and let $A \subseteq X$ be nonempty, open, and closed. Then $A = X$.

Proof. Since A is open and closed, A^c is open. Note that $A \cup A^c = X$ is a union of open sets. However, since X is connected, it follows that either A or A^c is not a proper subset of X by contrapositive, and thus $A = X$ since $A \neq \varnothing$. ∎

Theorem 4.1

Let Xbe connected, and let $f : X \to Y$ be continuous. Then $f(X)$ is connected.

Proof. If there exist $U, V \subseteq Y$ disjoint open sets such that $U \cup V = f(X)$, then the preimage $f^{-1}(U), f^{-1}(V)$ are disjoint and open in X, and

$$f^{-1}(U) \cup f^{-1}(V) = f^{-1}(U \cup V) = f^{-1}(f(X)) = X.$$

Since X is connected, one must be the entirety of X. Without loss of generality, let $f^{-1}(U) = X$. Then $f(X) \subseteq U$, and $V = \varnothing$. Therefore, there do not exist *proper empty* open sets $U, V \subseteq Y$ such that $U \cup V = f(X)$, and we are done. ∎

Theorem 4.2

Let $C \subseteq X$ with C connected, and let $C \subseteq A \subseteq \overline{C}$. Then A is connected. In particular, if A is connected, then $\overline{A}$ is connected.

Proof. Let there exist disjoint U_1, U_2 open in A such that $U_1 \cup U_2 = A$.Then $C \subseteq U_1 \cup U_2 \subseteq \overline{C}$. Then either $C \subseteq U_1$ or $C \subseteq U_2$. Without loss of generality, let $C \subseteq U_1$. We want to show $A = U_1$. The closue is simply $C \cup C'$, for C' the set of limit points of C, and thus $x \in A$ implies $x \in C$ or $x \in C'$. It suffices to show $C' \subseteq U$. Then for any $x \in C'$, for any open set U_x of x, there exists y such that $y \in C \cap U_1$. Therefore, such a y exists with $y \in U_1 \cap U_2$, a contradiction. (idk) ■

Theorem 4.3

Let $\{C_\alpha\}_{\alpha \in A}$ be a collection of connected subsets of a topological space X such that

$$\bigcap_{\alpha \in A} C_\alpha \neq \varnothing.$$

Then

$$\bigcup_{\alpha \in A} C_\alpha$$

is connected.

Proof. Suppose

$$\bigcup_{\alpha \in A} C_\alpha = U_1 \cup U_2$$

for U_1, U_2 open in X. Then for all $\alpha \in A$,

$$C_\alpha \subseteq \bigcup_{\alpha \in A} C_\alpha \subseteq U_1 \cup U_2.$$

Since

$$C_\alpha = (C_\alpha \cap U_1) \cup (C_\alpha \cap U_2),$$

and since C_α is connected, then either $C_\alpha \subseteq U_1$ or $C_\alpha \subseteq U_2$. WLOG, let $C_1 \subseteq U_1$. Since $\cap C_\alpha \neq \varnothing$, we have that

$$\bigcap_{\alpha \in A} C_\alpha \subseteq C_\alpha \subseteq U_1.$$

This can be done similarly for all $\alpha \in A$. If U_2 does not contain all C_α, then there must be some α' with $C_{\alpha'} \subseteq U_2$. However, since

$$\bigcap_{\alpha \in A} C_\alpha \neq \varnothing,$$

there must exist $x \in C_1 \cap C_2$, a contradiction. ■

Lecture 12: More on Connected

Tue 18 Mar 2025 11:18

We can define an equivalence relation $\sim$ on a topological space such that $x \sim y$ iff there exists a *connected* set $C \subseteq X$ such that $x, y \in C$. The only nontrivial property is transitivity, and this follows immediately from the previous theorem.

Definition 4.2: Locally Connected

A space X is locally connected if for all $x \in X$, there exists a connected open neighborhood $x \in C \subseteq X$.

Theorem 4.4

Given the equivalence relation above, each element of $X/\sim$ is connected in X. Moreover, for every $A \subseteq X$ connected, there exists $[x]$ with $A \subseteq [x]$, and each $[x]$ is closed. If X is locally connected, then each component is open.

I am too tired to follow this proof. We had homework that we had an extra week to do cuz of the break, but i got the due date wrong and only realized a day before it was due and there were these two problems that would probably take me a day to do each and i had to speedrun them super handwavingly because i had a day to do the entire homework and now im exhausted cuz i didnt sleep

Lecture 13: Connected

Thu 20 Mar 2025 11:07

Theorem 4.5

$\mathbb{R}$ is connected. In particular, any interval is connected.

Corollary 4.1

Any open set $U \subseteq \mathbb{R}$ is a disjoint union of open intervals.

Corollary 4.2

$\mathbb{R}^n$ is connected.

Corollary 4.3

S^2 is connected by steregraphical projection. More precisely, $\mathbb{CP}^1 \cong S^2 \cong \mathbb{R}^2$. Similarly, $S^1 \cong \mathbb{R}^2 \setminus \{0\} \cong \mathbb{R}_{>0}$ are connected.

Theorem 4.6

Let X be connected, and let $q : X \to Y$ be surjective and continuous. Then Y is connected. More precisely, if $q : X \to Y$ and the decomposition $q : X \to \operatorname{im} q$ is continous, then $\operatorname{im} q$ is connected.

Proof. Recall that all surjections are quotients up to isomorphism.

Let $U \cup V = Y$ with $U \cap V = \varnothing$. Then since q is continuous, $q^{-1}(U) \cup q^{-1}(V) = X$, but also $q^{-1}(U) \cap q^{-1}(V) = q^{-1}(U \cap V) = \varnothing$. Since X is connected, we must have either U or V empty. Without loss of generality, suppose U is empty. If $q^{-1}(U)$ is empty, then there exists no x such that $q(x) \in U$. Since q is surjective and Y has the quotient topology, we mus have U empty. ■

As an example, since the torus T^2 can be viewed as $[0,1] \times [0,1]/\sim$ for some relation $\sim$, it is connected in the quotient topology.

For the first time in topology, we can show spaces are *not* homeomorphic. For example, $\mathbb{R}^2$ is not homeomorphic to $\mathbb{R}$. Suppose for contradiction that $f : \mathbb{R}^2 \to \mathbb{R}$ is a homeomorphism, and choose $x \in \mathbb{R}^2$ such that $x \mapsto 0$. Then the restriction $\mathbb{R}^2 \setminus \{x\} \to \mathbb{R} \setminus \{0\}$ is still continuous, but clearly $\mathbb{R} \setminus \{0\}$ is not connected, whereas $\mathbb{R}^2 \setminus \{x\}$ is connected.

Theorem 4.7 (Intermediate Value Theorem)

Let X be connected, and let $f : X \to \mathbb{R}$ be connected. If $a < c < b$ with $a, b \in f(X)$, then $c \in f(X)$.

Proof. Suppose for contradiction that $c \notin f(X)$. Then $(-\infty, c), (c, \infty)$ are open and contain a, b respectively. These are open in X, and thus

$$f^{-1}(-\infty, c) \cup f^{-1}(c, \infty) = X \setminus f^{-1}(c) = X.$$

These sets are also trivially disjoint, and thus X is connected, a contradiction. ■

Theorem 4.8 (Brower Fixed Point Theorem)

Let $f : [-1, 1] \to [-1, 1]$ be continuous. Then there exxiss $x \in [-1, 1]$ such that $x = f(x)$.

Proof. If $f(-1) = -1$ or $f(1) = 1$, then we are done. Otherwise, assume $f(-1) > -1$ and $f(1) < 1$. Let $g(x) = f(x) - x$, which is continuous by composition of continuous maps. Then $g(-1) = f(-1) + 1 > 0$, and similarly $g(1) = f(1) - 1 < 0$. From the intermediate value theorem, there exists $x \in [-1, 1]$ such that $g(x) = 0$, and thus $f(x) = x$. ■

Definition 4.3: Path Connectedness

A topological space X is *path connected* if for every $x, y \in X$, there exists a continous map $\phi : [0, 1] \to X$ such that $\phi(0) = x$ and $\phi(1) = y$.

Lemma 4.1

If X is path connected, then X is connected.

Proof. Let $U, V \subseteq X$ such that $U \cap V = \varnothing$ and $U \cup V = X$. Suppose for contradiction that these sets are nonempty. Then there exists $u \in U$, $v \in V$, and thus there exists $\phi : [0, 1] \to X$ such that $\phi(0) = u$ and $\phi(1) = v$. Consider the preimages $\phi^{-1}(U), \phi^{-1}(V)$ as open sets in $[0, 1]$, and note that

$$\phi(U) \cup \phi^{-1}(V) = [0, 1], \qquad \phi^{-1}(U) \cap \phi^{-1}(V) = \varnothing.$$

Moreover, these sets are nonempty. Since $[0, 1]$ is connected, this is a contradiction. ■